

Pythagoras' Theorem & Trigonometry

Extra practice test 3 · 27 marks total

Name: _____ Date: _____ Score: _____ / 27

Section A — Pythagoras' theorem

A1. A square has a diagonal of 18 cm. Find the length of one side, to 1 decimal place.

[2 marks]

A2. An equilateral triangle has sides of length 16 cm. Find its perpendicular height, to 1 decimal place.

[3 marks]

A3. A triangle has sides of length 11 cm, 60 cm and 61 cm. Determine, showing your working, whether this triangle is right-angled.

[2 marks]

Section B — SOH CAH TOA basics

B1. True or False: in a right-angled triangle, increasing the angle θ (while keeping the hypotenuse fixed) increases the length of the side opposite θ . Explain your answer.

[1 mark]

B2. Without using a calculator, state whether $\tan 80^\circ$ is greater than or less than 1. Explain your reasoning.

[1 mark]

B3. Calculate $\sin 70^\circ$ and $\cos 70^\circ$ (to 2 decimal places) and state which is larger.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has a hypotenuse of 22 cm and an angle of 40° . Find the length of the side adjacent to the 40° angle, to 1 decimal place.

[3 marks]

C2. A vertical flagpole casts a shadow 9 m long when the sun's angle of elevation is 51° . Find the height of the flagpole, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 24° and an opposite side of 5 cm. Find the length of the adjacent side, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A road climbs 18 m over a horizontal distance of 150 m. Find the angle of the slope, to 1 decimal place.

[2 marks]

D2. A right-angled triangle has a hypotenuse of 19 cm and one shorter side of 11 cm. Find the angle between the hypotenuse and the 11 cm side, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A window cleaner's ladder must make an angle of at least 65° with the ground to be considered safe. The ladder is 4 m long and its foot is placed 1.5 m from the wall. Calculate the angle the ladder makes with the ground and state whether it meets the safety requirement.

[2 marks]

E2. A ship sails from port P for 15 km due east to point Q, then turns and sails 9 km due north to point R. (a) Find the direct distance from P to R, to 1 decimal place. (b) Find the angle between the direct route PR and the initial direction PQ (due east), to 1 decimal place.

[3 marks]